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# Grocery product detection and recognition

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ABSTRACT

Object detection and recognition are challenging computer vision tasks receiving great attention due to the large number of applications. This work focuses on the detection/recognition of products in supermarket shelves; this framework has a number of practical applications such as providing additional product/price information to the user or guiding visually impaired customers during shopping. The automatic creation of planograms (i.e., actual layout of products on shelves) is also useful for commercial analysis and management of large stores.

Although in many object detection/recognition contexts it can be assumed that training images are representative of the real operational conditions, in our scenario such assumption is not realistic because the only training images available are acquired in well-controlled conditions. This gap between the training and test data makes the object detection and recognition tasks far more complex and requires very robust techniques. In this paper we prove that good results can be obtained by exploiting color and texture information in a multi-stage process: pre-selection, fine-selection and post processing. For fine-selection we compared a classical Bag of Words technique with a more recent Deep Neural Networks approach and found interesting outcomes. Extensive experiments on datasets of varying complexity are discussed to highlight the main issues characterizing this problem, and to guide toward the practical development of a real application.

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## 1. Introduction

Detection and recognition of objects in a scene play an important role in many real applications. The recent growth of mobile devices equipped with high quality cameras enables applications supporting everyday life activities in different contexts. This work focuses on the specific problem of detecting/recognizing grocery products from shelves in a supermarket; automatic product detection can be useful for different purposes such as providing additional product/price information to the user or guiding visually impaired customers during shopping. Object detection/recognition in grocery scenarios is very complex because object appearance is highly variable due to relevant changes in scale, pose, viewpoint, lighting conditions and occlusion. Additional peculiarities characterize the grocery product domain (see Fig. 1). On the one hand, different products can be very similar and only minor packaging details allow to discriminate them (e.g. the background color or a slightly different text) (Fig. 1(a)): this often happens when different flavors of the same product are available. On the other hand, the packaging of a given product can vary over time; for instance special offers advertisement is sometimes reported on the package (see Fig. 1(b)). It is worth noting that categorizing products in macro-categories (e.g. detergent, pasta, chips, etc.) is a relatively easier task than detecting/recognizing specific products, especially when real-time processing is needed.

Another peculiarity of this scenario is that usually acquiring training data in real environments is impracticable and the training phase, aimed at populating a reference database with representations for the products of interest, relies on few images per product acquired off-line in ideal conditions, often taking pictures of the same product under different poses in a rotating platform.

In this work we propose a novel approach for product detection and recognition from shelves. The proposed approach consists of three steps, outlined in Fig. 2: *pre-selection*, aimed at selecting the initial set of candidate windows on the basis of the joint information obtained from corners position and color distribution, *fine-selection* where more robust features are applied for candidate selection and *post-processing* where multiple detections of the same object are clustered to produce the final result.

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**Fig. 1.** In (a) different products are shown (small inter-class variability), in (b) the same product with two different packages is reported (high intra-class variability).

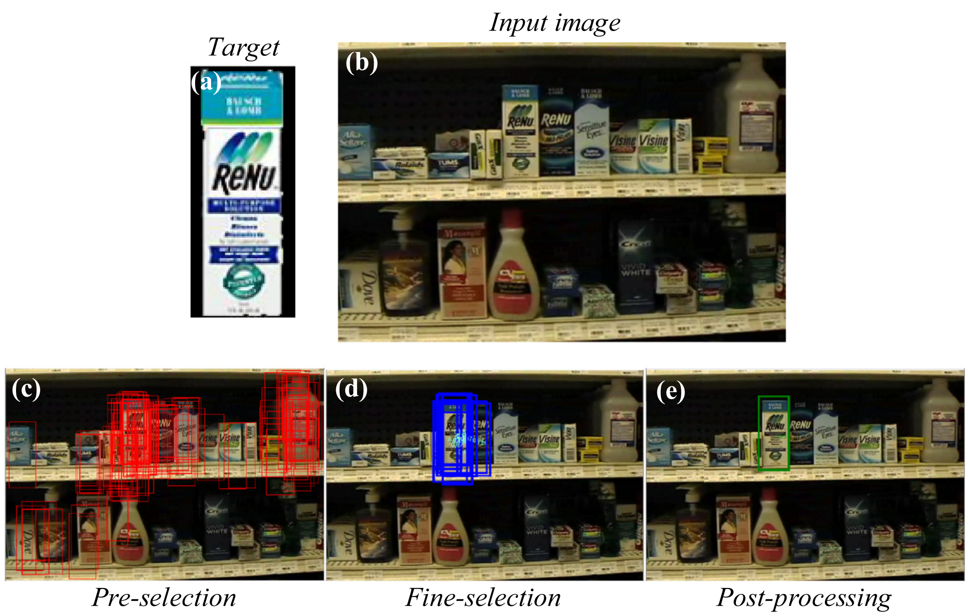


**Fig. 2.** The main steps of our product detection algorithm. Given a target product (a) and an input image (b), the candidate windows are selected on the basis of corner position and color information (c), then the fine-selection stage discards the candidates which are not compatible with the target product (d) and the final result (e) is obtained by consolidating multiple detections of the same product instance.

The main contributions of our work are:

– a novel fast approach for candidate pre-selection based on object corners and color information that overcomes the inefficiency of exhaustive sliding window approaches and, in this scenario, outperforms candidate selection via selective search (Uijlings, Van de Sande, Gevers, & Smeulders, 2013);
– a robust approach for grocery object detection and recognition where two different representations (Bag of Words and Deep Neural Network) are proposed and compared;
– the introduction of an accurate post-processing step, used to combine multiple detection of the same object and to determine the final detection output. Such step, often overlooked or briefly described in the literature, proved to be crucial to maximize performance, especially when accuracy is evaluated using very restrictive criteria as most recent benchmarks impose.
– a deep experimental evaluation on the complex Grozi-120 public dataset (Merler, Galleguillos, & Belongie, 2007), including a favorable comparison with the methods presented in (Merler et al., 2007).

The paper is organized as follows: in Section 2 the related literature is reviewed; in Section 3 we present the pre-selection procedure; Section 4 describes the fine-selection process analyzing two possible feature extraction techniques (based on multi-resolution BoW features and DNN features). The post-processing technique is discussed in Section 5, Section 6 presents the experiments and, finally, Section 7 draws some conclusions.

## 2. Related works

The generic problem of object detection has been widely investigated by the research community and several solutions have been proposed in the literature: see Li, Wang, Tian, and Ding (2015), Shantaiya, Verma, and Mehta (2013), Parekh, Thakore, and Jaliya (2014) and Hemalatha, Muruganand, and Maheswaran (2014) for recent surveys on this topic. Even if a general purpose object detection algorithm would be highly desirable, the extreme variability that characterizes real objects makes it very difficult to find a general approach; in fact, domain-specific peculiarities can improve object representation. For instance, a lot of efforts have been devoted to the study of face detection techniques and effective solutions have been provided by the research community (Yang, Kriegman, & Ahuja, 2002; Zhang & Zhang, 2010), even if the problem is still challenging in uncontrolled scenarios (e.g. Zafeiriou, Zhang, & Zhang, 2015).

The problem of on-the-shelf product detection has been addressed by some works in the literature. In (Tsai et al., 2010) a product recognition system for mobile devices is presented, with particular focus on the application design and only marginal discussion on the recognition problem.

**Table 1**
Comparison of the existing techniques in terms of topic addressed, features adopted, pros and cons.

| Paper | Topic | Features | Pros | Cons |
|---|---|---|---|---|
| Tsai et al. (2010) | Product recognition | cHoG | Focus on client-server system architecture with accurate processing time analysis | No object detection Limited analysis of recognition accuracy |
| George and Floerkemeier (2014) | Product categorization | SIFT | New annotated database (*Grocery Products*) | Only macro category detection |
| Varol and Kuzu (2014) | Product detection and recognition | SIFT, HSV (color) | Good detection performance | Shelf-based (not general); specific for tobacco packages |
| Merler et al. (2007) | Product detection and recognition | Color histograms, SIFT, Haar-like features | New annotated database (*Grozi-120*) | Limited accuracy |
| Winlock et al. (2010) | Product detection | SURF, color histograms | Real time application (exploiting video information) | Focus on efficiency, limited analysis of accuracy |

In George and Floerkemeier (2014) and George, Mircic, Soros, Floerkemeier, and Mattern (2015) the problem of product class recognition is studied. The authors collected a database (*GroceryProducts*) with related ground truth with the objective of classifying products into broad classes such as Food/Bakery, Food/Biscuits, etc.; each class contains different products and a single training image per product. As previously noted, the problem addressed in these works is slightly different from that treated in this paper since our classes coincide with specific products and not with product macro-categories.

The authors of Varol and Kuzu (2014) consider the problem of retail product recognition on grocery shelf images. Firstly, the objects of interest are detected using an ad-hoc cascade classifier trained with HOG features to identify a specific class of objects (e.g. tobacco packages in the paper); subsequently SIFT features are used for identifying the specific product. The knowledge about the context of the shelf structure is exploited to constrain the detection; such assumptions, feasible for specific classes of products, have not a general validity since different categories of products have different layouts in the store, and not always shelves can be detected.

Finally, in Merler et al. (2007), the closest approach to our work, the problem of product detection and recognition in real environments, starting from “in vitro” training images, is addressed. Different feature extraction approaches are compared both in terms of recognition and detection accuracy: color histograms, SIFT and Haar-like features. Besides this interesting comparison, the paper represents an important reference since it introduces the *GroZi-120* dataset, one of the few existing benchmarks for the problem of product detection in real environments. The database consists of 120 products of different categories; the proposed testing protocol precisely identifies training and test images and constitutes a useful baseline for performance comparison. GroZi-120 dataset is also used in Winlock, Christiansen, and Belongie (2010) where the *ShelfScanner* system is proposed to support visually impaired users to shop at a grocery store. *ShelfScanner* exploits video information to efficiently estimate the position of the objects of interest in the incoming frames on the basis of the optical flow. In particular, a sort of “mosaic” is progressively built to distinguish the regions which already appeared in the previous frames from the new ones that must be processed. A combination of SURF features and color histograms is implemented for product detection.

A comparative analysis of the state-of-the-art is provided in Table 1 where the main characteristics of the existing algorithms are summarized.

## 3. Candidate pre-selection

The initial step of the pre-selection procedure is the segmentation of the foreground (shelves and products) from the background (see Fig. 3a) which is typically characterized by dark colors; this segmentation can be simply achieved by a fixed-threshold binarization. The foreground map is then used to enforce a simple constrain: all selected windows must cover a minimum percentage of foreground.

Several approaches have been proposed in the literature for candidate pre-selection. One of the most used techniques is the sliding-window scan (Dalal & Triggs, 2005; Laptev, 2006; Papageorgiou & Poggio, 2000; Sabzmeydani & Mori, 2007; Shashua, Gdalyahu, & Hayun, 2004; Tuzel, Porikli, & Meer, 2007; Viola & Jones, 2004); such approach produces a huge number of candidates and a very fast technique has to be adopted to quickly filter out the undesired windows in real time.

Recently some interesting techniques have been proposed to increase the efficiency of the candidate selection procedure; a relevant representative is the Selective Search approach presented in Uijlings et al. (2013) where a hierarchical segmentation is adopted to identify possible objects of interest at different resolutions. Such techniques generally perform well on natural scenes but fail to address the peculiarities of the grocery images where no background regions can be identified (except for some very limited dark areas not covered by the shelf products) and the product packaging usually presents a non-uniform color thus making unfeasible a candidate selection based on color and texture homogeneity. Our preliminary experiments confirmed that Selective Search is not well suited for this specific application.

The nature of our target objects suggested to start searching candidates from the image corners, identified using the well-known Harris corner detector (Harris & Stephens 1988). In fact, most of the grocery products have a quite regular shape and several corners originate in the boundary between the product and the shelf/background. Of course many other corners will be detected inside the product due to the characteristics of the packaging which includes labels, drawing and inscriptions (see Fig. 3b).

For this reason, the corners represent only a starting point for further evaluations based on the color distribution. In particular, each corner suggests four candidate windows, as shown in Fig. 4. Each of the four windows is then checked against the following two conditions:

- the gray-scale variance in the window must be higher than a prefixed threshold (to discard too uniform regions such as empty shelves or walls);
- the percentage of foreground pixels is over a threshold.

If both conditions are fulfilled, the 3D color histogram of the window, computed in the YCbCr color space (Chai & Ngan, 1999), is compared against the reference histogram of the target product (pre-computed and stored in the database) by using a simple histogram intersection metric. The significant differences between training and test images make necessary a color normalization before comparison. In the literature many normalization techniques have been proposed to improve the visual quality of


**Fig. 3.** The dark area behind the products on the shelves can be easily isolated by a simple image binarization (a). The corners detected in the image represent the starting point for candidate window detection (b).


**Fig. 4.** Each rectangle represents a possible candidate originated from the central (white-filled) corner. Of the four candidates, only the green one satisfies all the constraints (foreground presence and color distribution) and will thus be passed to the fine-selection step.

single images; for instance the adaptive algorithms such as CLAHE (Zuiderveld, 1994) are particularly well-suited to deal with regions characterized by different illuminations (very likely to occur in natural scenes). In our specific domain, the illumination is usually rather uniform within a single (training or test) image, while it can substantially differ between training and test. We therefore adopted a simple pre-processing which operates on individual color channels by rescaling the intensities to obtained a fixed mean and variance (mean 0 and standard deviation 3).

A visual example of the normalization effects obtained using the adaptive general purpose CLAHE approach (Zuiderveld, 1994) and our normalization is given in Fig. 5. As clearly visible, CLAHE offers a better result from the visual point of view, but does not make training and test comparable; our normalization produces a more “washed out” effect, but makes the two images more similar. The experimental results reported in Section 6.1, where the two normalization techniques are compared, confirm such behavior.

Among the candidate windows with color intersection higher than a threshold the best one, i.e. the one whose histogram has a higher intersection with the target product, is chosen as candidate for further processing.

By proper optimizations (e.g. computation of the integral image for intensities variance and percentage of foreground pixels), the above conditions can be evaluated very quickly. An example of the selection procedure is show in Fig. 4: the red window is discarded since the covered foreground area is too small, the yellow windows do not fulfill the constraint related to the color histogram while the green window passes the selection. To take into account scale changes due to the product distance from the camera, the candidate windows for each corner are evaluated at different scales. The pseudo-code of the Pre-Selection procedure is reported in Fig. 6.

## 4. Fine selection

The *fine-selection* stage processes the candidates resulting from the *pre-selection* to further select them on the basis of more robust features. In our application domain the color information is certainly very relevant and useful to discriminate different products or different versions (typically flavors) of the same product; in the proposed approach the discriminative capabilities of the color features are exploited in the pre-selection stage (see Section 3) to quickly identify the windows of interest, thus limiting the complexity of the classification stage. While color information could have been used in the fine-selection stage too, mainly for efficiency reasons we used grayscale features and we combined the scores obtained from fine-selection (related to texture) with those computed during pre-selection (i.e. related to color information).

For each product in the database, more training images are available; the reference model is then a set of $d$-dimensional vectors (one for each training image): $TS = \{\mathbf{ts}_1,\ \mathbf{ts}_2, \ldots, \mathbf{ts}_N\}$. Two different feature extraction techniques are presented in the following subsections. Let $\mathbf{f}$ be the feature vector associated to a candidate $\mathbf{r}_i = (\mathbf{w}_i,\ s_i)$ coming from the pre-selection stage, then its similarity score with a model $\mathbf{ts}_i$, is computed as:

$$score = \left(s_f \times s_m \times s_i\right)/sThr$$

where:

- $s_i$ is the score obtained in the pre-processing stage
- $s_f$ represents the best match between $\mathbf{f}$ and $TS$:

$$s_f(\mathbf{f},\ TS) = \max_{i=1,\ldots,N} sim(\mathbf{f},\ \mathbf{ts}_i)$$


**Fig. 5.** Normalization effects produced by CLAHE (b) and our normalization (c) on a pair of training-test images (a).

```
Input C = set of corners resulting from Harris corner detection
Input SF = {sf_1, sf_2, .., sf_t}, is the set of t scales considered
Input TP = target product to be detected
Output R = candidate windows, each defined by a rectangle w and a score s

PRE-SELECTION (C, SF, TP)
R = ∅
h ← Load the pre-calculated 3D color histogram of the target product TP
for each corner c_i = (x_i, y_i) ∈ C
    for each sf ∈ SF
        W = {w_j, j = 1,..,4} ← Compute the candidate windows for corner c_i at scale sf
        s_best = 0
        for each candidate w_j ∈ W
            f ← Compute foreground percentage in window w_j
            if f > fThr then
                v ← Compute gray-scale variance in window w_j
                if v > vThr then
                    h_j ← Compute color histogram of window w_j
                    s ← HISTOGRAMINTERSECTION(h_j, h)
                    if s > sThr ∧ s > s_best then
                      s_best ← s, w_best ← w_j
                end if
            end if
        end for
        if s_best > 0 then
            R ← R ∪ {w_best, s_best}
        end if
    end for
end for
return R
```

**Fig. 6.** Pseudo-code of the PRE-SELECTION algorithm.

– $s_m$ considers the number of matches between $\mathbf{f}$ and $TS$:
–

$$s_m = \frac{\#M}{\#TS} \text{ where } M = \{ts_i \in TS \ : \ sim(\mathbf{f},\ \mathbf{ts}_i) > sThr\}$$

where $sim(\mathbf{f}, \mathbf{ts}_i)$ depends on the features representation (see following subsections); the threshold value $sThr$ is product-specific and is determined on a separate validation set.

### 4.1. Bag of words representation

The Bag of visual Words (BoW) model (Zhang & Marszalek, 2006) is an approach inspired by well-known text retrieval techniques. The idea is to represent an object as a histogram of occurrences of the “words” contained in a reference dictionary. In the context of visual object retrieval, the concepts of *visual words* and *visual dictionary* have to be properly defined; in particular, the *visual words* are typically associated to local features extracted from a large set of training images and the *dictionary* is computed through quantization (i.e. clustering) of the training words. The adoption of the BoW model in this work is motivated by the possibility of extracting a compact fixed-length descriptor from the product images: this peculiarity is useful in our application because it allows to better deal with product size changes due to the camera distance.

The local features we used to encode the visual words are the SURF descriptors (Bay, Ess, Tuytelaars, & Van Gool, 2008) sampled uniformly over the gray-scale version of the considered image patch; a dense sampling (honeyrate structure with step $p$) is here preferred with respect to a saliency based key point detection technique for efficiency reasons.

A reference dictionary is built by applying K-means clustering (Fukunaga, 1990) to the set of SURF descriptors extracted from the

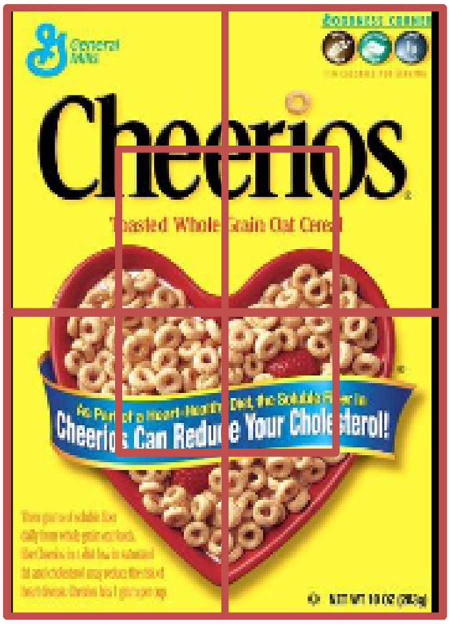


**Fig. 7.** The image partitioning approach adopted for local feature extraction.

training images of all the target products. We denote with $s$ the number of clusters determined (i.e. the dictionary size) and with $n$ the dimensionality of a single word (i.e. the dimensionality of the SURF descriptors adopted, typically 64 or 128 elements).

Let $V = \{v_1, v_2, \ldots, v_s\}$ be the dictionary computed in the training phase. Given a product image $I$ to be represented, a set of SURF descriptors $D = \{d_1,\ d_2, \ldots, d_m\}$ is computed over $I$; then, the feature vector $\mathbf{f}$ representative of image $I$ consists of the histogram of occurrences in $D$ of the words belonging to $V$. Being our features continuous values, a precise correspondence between the words in the dictionary and the descriptors is very unlikely; therefore when computing the histogram each descriptor $d_j$ is associated to the closest word $v_{i^*}$ in the dictionary:

$$i^* = \underset{i}{argmin} \|d_j - v_i\|.$$

Since the histogram is a statistical descriptor globally summarizing the frequency of some basic “patterns” over the whole image, it does not encode information about the spatial information in the image. To avoid completely losing this information we divide the image into $r$ overlapping regions as shown in Fig. 7, encode each region separately (with his own histogram) and concatenate the features into a $d = r \times s$ dimensional vector. The proposed region overlapping implicitly gives more strength to the central region w.r.t. the surrounding ones which are often incomplete or noisy (e.g. they could contain parts of other neighboring products in the shelf image).

To speed-up the fine-selection step a two-level approach is adopted: at the first level we work at low-resolution and quickly pass over candidates with high similarity with respect to the target product; if the low-resolution response is not totally convincing we re-process the candidate windows at higher resolution. Different resolutions are obtained by varying the sampling density of the local descriptors.

To compute the similarity score between a feature vector $\mathbf{f}$ and the reference model $TS = \{\mathbf{ts}_1,\ \mathbf{ts}_2, \ldots, \mathbf{ts}_N\}$ we used a histogram intersection metric (Swain & Ballard, 1991). In particular, given two $d$-dimensional histograms $\mathbf{f}$ and $\mathbf{ts}_i$, their similarity (i.e. intersection) is defined as follows:

$$sim(\mathbf{f},\ \mathbf{ts}_i) := s_{hist}(\mathbf{f},\ \mathbf{ts}_i) = \sum_{k=1}^{d} \min\left(f_k, ts_{ik}\right)$$

In Fig. 8 we reported the pseudo-code of the FineSelection procedure with BoW features.

*4.2. Deep neural network features*

In many object detection and recognition tasks, the adoption of CNN (Convolutional Neural Networks), a special deep neural network (DNN), has recently provided surprising results (LeCun, Bengio, & Hinton, 2015). In this work we evaluate their effectiveness for grocery product detection.

It is well-know that large deep networks need huge amount of training data to learn complex invariances, and our in-vitro training data (even if extended with data augmentation techniques) is not enough to train a new CNN from scratch. Directly using CNN for product detection (i.e. without candidate preselection) is still more critical in case of scarcity of training data. In fact state-of-the-art detectors such as Faster R-CNN (Ren, He, Girshick, & Sun, 2015) require training a classifier (CNN) to classify regions and a regressor (RPN) to generate region proposal. Both require huge amount of labeled data: classes for CNN, and bounding boxes for RPN. Fortunately, several object recognition/detection problems have been effectively addressed by using the so-called transfer learning approach, i.e. exploiting a pre-trained CNN as feature extractor. In fact, several works in the literature, show that CNN intermediate-layer features are general enough to be applicable to different problems (Yosinski, Clune, Bengio, & Lipson, 2014).

In this work, we used the AlexNet network (Krizhevsky, Sutskever, & Hinton, 2012) which has been pre-trained on more than 1 million images (and 1000 classes) from ILSVRC12 subset of ImageNet (Russakovsky et al., 2015).

Although AlexNet input is an RGB image, in our experiments we observed negligible advantages by using color information (after a color-based preselection); therefore, for uniformity with BoW here too we used gray-scale input images, rescaled to $227 \times 227$ pixels (AlexNet input size). The intermediate features are extracted at level “fc6”, i.e. the first fully connected level. The output of the network is a feature vector $\mathbf{f}$ of size 4096; the simple Euclidean distance is used to compare two feature vectors; since a similarity score is assumed in the algorithm rather than a distance value, the score is computed as follows:

$$sim(\mathbf{f},\ \mathbf{ts}_i) = \frac{1}{\sqrt{\sum_{j=1}^{4096} \left(f_j - ts_{ij}\right)^2}}$$

The *fine-detection* procedure is very simple in this case. In fact, the multi-resolution sampling (that in case of SURF features was related to the keypoints density) is not applicable here and all the candidates with a score higher than the predefined threshold are retained.

## 5. Post-processing

In object detection a final post-processing is typically needed because the previous steps could produce multiple overlapped detections (window candidates) for the same instance of a product; such detections must be combined into a single output, otherwise multiple occurrences of the same product would be considered false positives. This stage is rather critical for the overall detection performance, particularly in the presence of restrictive evaluation protocols where a detection is considered correct only in the presence of a large overlap between the detected object and the ground truth. Traditional post-processing techniques such as

```
Input R = candidate windows set
Input TS_l = reference model of the target product at low sampling density
Input sThr_l: similarity threshold associated to TS_l
Input TS_h: reference model of the target product at high sampling density
Input sThr_h: similarity threshold associated to TS_h
Output R_new = refined candidate windows

FINESELECTION (R, TS_l, sThr_l, TS_h, sThr_h)
R_new ← ∅
for each candidate r_i = (w_i, s_i) ∈ R
    f_l ← Extract first level features from w_i
    s_fl ← max_{i=1,...,N} sim(f_l, ts_{l_i}), s_ml ← #matches(f_l, TS_l)/#TS_l

    if s_fl > sThr_l then
        s' ← (s_fl × s_i × s_ml) / sThr_l    // consolidate with color-based similarity from pre-selection
        R_new ← R_new ∪ (w_i, s')
    else if s_fl > (0.9 · sThr_l) then
        f_h ← Extract second level features from w_i
        s_fh ← max_{i=1,...,N} sim(f_h, ts_{h_i}), s_mh ← #matches(f_h, TS_h)/#TS_h

        if s_fh > sThr_h then
            s' ← (s_fh × s_i × s_mh) / sThr_h
            R_new ← R_new ∪ (w_i, s')
        end if
    end if
end for
return R_new
```

**Fig. 8.** Pseudo-code of the FINE-SELECTION algorithm with BoW features.

the well-known Non-Maxima Suppression (Neubeck & Van Gool, 2006) retain locally the best candidate, but in this way the object shape is not always well approximated. The rationale behind our proposal is to determine the final object, i.e. its shape and score, taking into account the contribution of all the local candidates. An iterative clustering algorithm has been designed that exploits the window score, position and size to create clusters of candidates and to determine a representative candidate for each cluster. The output of this stage is a set $CL = \{Cl_1, Cl_2, \ldots, Cl_m\}$ of clusters representing groups of candidate windows, where each cluster $Cl_i = (w_i, s_i)$ is represented by an image window $w_i$ (top-left corner and size) and a score $s_i$.

To compute the clusters the candidates in $R_{new}$ (i.e., coming from the previous stage) are considered in decreasing order of score; a first cluster is initialized with the highest score candidate $r_* = (w_*, s_*)$:

$$\mathrm{C} = \{Cl_1\} \text{ where } Cl_1 = (\mathbf{w}_*, s_*)$$

For each subsequent candidate $r_j = (\mathbf{w}_j, s_j)$ in $R_{new}$ the algorithm evaluates its compatibility with the existing clusters in $C$; in particular $r_j$ is assigned to $Cl_i = (w_i, s_i)$ if

$$\mathbf{w}_j \cap \mathbf{w}_i \geq \mathbf{w}_j \cap \mathbf{w}_k \ \forall\ k \neq i \text{ and } \mathbf{w}_j \cap \mathbf{w}_i > \frac{area(\mathbf{w}_j)}{2}$$

If $r_j$ is assigned to $Cl_i$, the cluster is updated to take into account the contribution of the new entry: in particular, the cluster window is computed as the average of the cluster members windows, weighted by their scores; the cluster score is updated by adding the score of the new entry. If a candidate cannot be assigned to any of the existing clusters, a new cluster is created.

A final threshold *clThr* is applied to eliminate the clusters whose score is too small with respect to the maximum score of the identified clusters. This choice is based on the observation that the score of each cluster is somehow proportional to the size of the detected product since in the presence of a big product several candidates will be detected, thus contributing to increase the score. For this reason, we can assume that high cluster scores denote the presence of big products in the image; in this case small scores are likely to represent outliers and can thus be eliminated. The final set of clusters is then:

$$\mathrm{C} = \cup\, Cl_i \mid s_i > clThr$$

A comparison between the proposed post-processing and a more conventional Non-Maxima Suppression (Neubeck & Van Gool, 2006) is given in Section 6.2.

## 6. Experiments

Extensive experiments have been conducted to validate the proposed method. We first evaluated the effectiveness of the proposed feature extraction algorithm; then a second set of tests was carried out to validate the whole detection process, and to evaluate the performance according to well-established metrics.

A very limited number of datasets is publicly available for this specific application. Two datasets, characterized by a different degree of complexity, have been used in our testing:


**Fig. 9.** Products selected from Grocery Products DB (GP-20 DB).

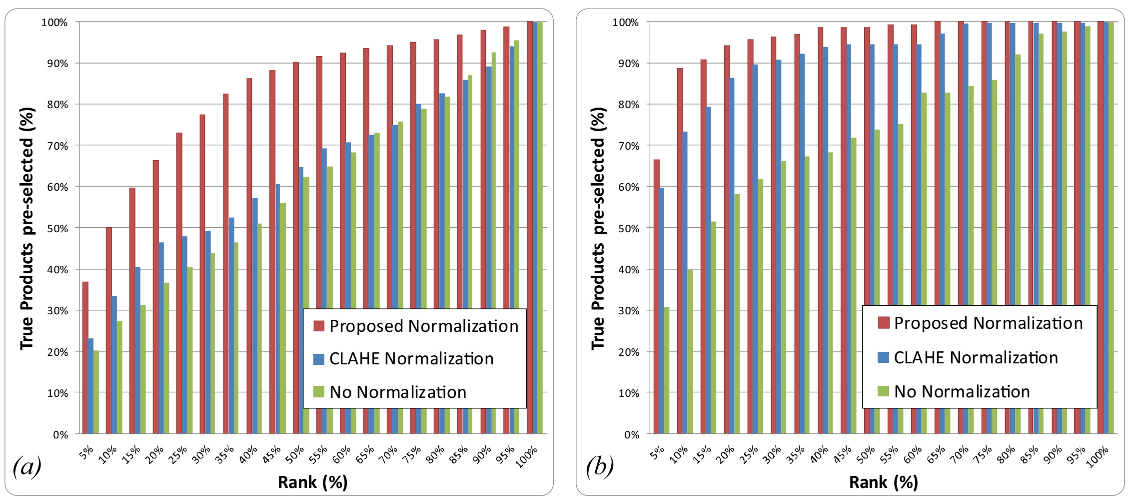


**Fig. 10.** The cumulative product ranking curve for the Grozi-120 db (a) and GP-20 db (b); the results without color normalization and with normalization (Proposed and CLAHE) are given.

**Table 2**
Area Under ROC curve (AUROC) for tests on Grozi-120 and GP-20 dbs measured for the fine-selection approach (Section 4) with and without the color-based pre-selection step.

| | | Grozi-120 | GP-20 |
|---|---|---|---|
| BoW | Pre-selection | 0.9482 | 0.9964 |
| | No pre-selection | 0.9485 | 0.9935 |
| DNN | Pre-selection | 0.9687 | 0.9997 |
| | No pre-selection | 0.9728 | 0.9988 |

- The first one is a public database of 120 groceries, GroZi-120 (Merler et al., 2007) available at http://grozi.calit2.net. The database contains objects of different colors, size, shape and stiffness. Each product was acquired both in vitro modality (i.e., in ideal laboratory conditions of brightness or positioning), and in situ modality (i.e., cropped from videos coming from real environments under noticeable variations in terms of scale, illumination and poses, see Fig. 4). This dataset was realized with the specific aim of assessing algorithm performance on real (in situ) images when the system is trained using in vitro images only, thus evaluating the generalization capabilities and the robustness of the algorithm. The testing protocol given in (Merler et al., 2007) has been adopted in order to obtain comparable results: for the detection tests a total of 13,200 images processed by selecting 110 images for each product (10 test images taken from videos containing the product and 100 images not containing any product of interest).
- The second database, called GP-20 in the following, consists of 20 groceries selected from the "Grocery Products" dataset (George & Floerkemeier, 2014). The "Grocery Products" dataset, available at http://people.inf.ethz.ch/mageorge, is a collection of 3235 products clustered in 27 classes. Each product is represented by only one training image, taken in ideal conditions with a white background. It contains 680 annotated test images taken in real scenario with products of all subcategories in the "Food" category. The database was proposed to detect categories of products rather than specific products and the annotations of test images are not suited for our object detection purpose; each element of the ground truth, in fact, contains multiple instances of products belonging to the same subcategory (see Fig. 3 of George & Floerkemeier, 2014). For this reason, starting from this dataset, GP-20 has been created by selecting 20 groceries from the training set (see Fig. 9) and manually defining a ground truth analogous to that provided in the Grozi-120 dataset. The test set used for detection consists of 61 test images of shelves containing the selected products and 10 images without any product of interest.

### *6.1. Evaluation of the color and texture features*

The recognition experiment proposed in the protocol of the Grozi-120 database is particularly well-suited to evaluate the effectiveness of the features extracted from the product images; in fact, it allows isolating the contribution of the features from the rest of the detection process. More in details, 10 in situ images for each product, extracted from the testing frames according to the ground truth information, are used for testing; each image represents a single product and the algorithm has to perform a 1-N recognition to output the most similar product in the database. The recognition task is accomplished by making a pre-selection of the candidate products based on the color histogram similarity (see Section 3); the candidates are then compared to the test product using the proposed features (BoW and DNN) and the most similar one is

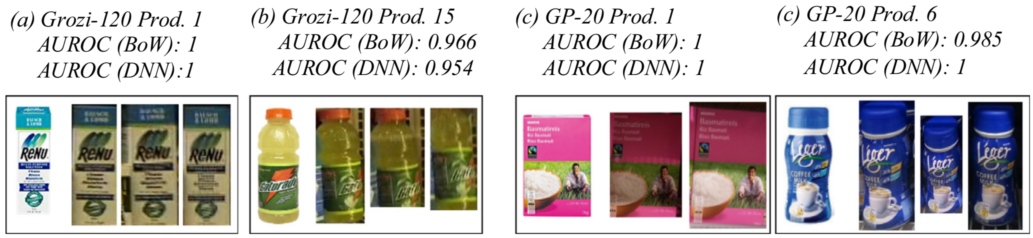


**Fig. 11.** Some examples from Grozi-120 (product 1 (a) and product 15 (b)) and GP-20 (product 1 (c) and product 6 (d)); for each product one training sample, some testing images and the AUROC value are reported.

**Fig. 12.** Detection results obtained with the proposed post-processing (left) and Non-Maxima Suppression (right) on the same test image.


**Fig. 13.** Example of detection of (in vitro) products (top) in testing images (bottom) from the Grozi-120 dataset.

**Table 3**
Grozi-120 db – AUROC values for different versions of the BoW-based feature extraction approach: single resolution with $p = \{3,\ 6,\ 12\}$ with $s = \{200,\ 1000\}$ and multiresolution (6-3) and (12-6) with $s = 200$.

| $p = 3$ | | $p = 6$ | | $p = 12$ | | *Multi* 6-3 *s=200* | *Multi* 12-6 *s=200* |
|---|---|---|---|---|---|---|---|
| *s=200* | $s = 1000$ | *s=200* | $s = 1000$ | *s=200* | $s = 1000$ | | |
| 0.9492 | 0.9557 | 0.9470 | 0.9339 | 0.8085 | 0.8098 | 0.9482 | 0.8707 |

**Table 4**
Detection performance of the proposed approach (BoW and DNN features), compared to the state-of-the-art results for Grozi-120 db. For comparisons with the results obtained in (Merler et al., 2007) it can be easily noted that, in the discrete metric, our Recall indicator corresponds to the TP(%) mentioned in, while the definition of FP(%) mentioned in (Merler et al., 2007) is not clear.

| *Algorithm* | *Post-processing* | *Continuous metric* | | *Discrete metric* | | |
|---|---|---|---|---|---|---|
| | | *Overall recall* | *Overall precision* | *Recall* | *Precision* | $FP_{avg}$ |
| ***Proposed approach BoW*** | *Proposed post-processing* | 41.8 | 39.2 | 46.3 | 45.7 | 1.0 |
| | *Non-maxima suppression* | 36.7 | 39.2 | 42.5 | 45.8 | 0.16 |
| ***Proposed approach DNN*** | *Proposed post-processing* | 44.4 | 37.5 | 52.7 | 45.2 | 0.18 |
| | *Non-maxima suppression* | 34.7 | 37.2 | 39.1 | 45.7 | 0.06 |
| ***CHM*** | | 15.0 | 17.0 | 18.0 | / | / |
| ***SIFT*** | | 20.0 | 18.0 | 22.0 | / | / |
| ***ADA*** | | 15.0 | 17.0 | 18.0 | / | / |

**Table 5**
Detection performance of the proposed approach on GP-20 db by using the continuous and discrete metric.

| | | *Continuous metric* | | *Discrete metric* | | |
|---|---|---|---|---|---|---|
| | | *Overall recall* | *Overall precision* | *Recall* | *Precision* | $FP_{avg}$ |
| ***Proposed BoW*** | *Proposed post-processing* | **65.4** | **73.7** | **76.5** | **77.7** | **0.94** |
| | *Non-max supp* | 58.5 | 74.2 | 70.2 | 76.6 | 0.22 |
| ***Proposed DNN*** | *Proposed post-processing* | 54.7 | 73.9 | 73.6 | 73.1 | 2.02 |
| | *Non-max supp* | 57.2 | 69.9 | 70.9 | 72.1 | 1.18 |


**Fig. 14.** Example of detection for a specific (in vitro) product (top) and a related testing image (bottom) from the GP-20 dataset.

selected. For each product, a ROC curve is computed starting from the genuine and impostor scores and the overall ROC curve is obtained by averaging the single product curves. As a compact indicator we report in the following the Area Under ROC curve (AUROC).

A first experiment focuses on pre-selection: for each test product, the ranking of the training products based on their 3D color Histogram similarity scores has been computed, and the percentage of true products pre-selected is extracted. In Fig. 10 the average cumulative product ranking curve is shown for both databases; the results obtained without color normalization are compared to those obtained with our normalization and with CLAHE (see Section 3). The curve obtained for the GP-20 db shows that 90% of the true products are ranked in the first 15% thus suggesting that test and training images are rather similar in terms of color features; for the Grozi-120 db only the 59% of true products are ranked in the first 15% due to a weaker color similarity between test and training samples. The results also show that the color normalization if fundamental, particularly in the Grozi-120 dataset where the color gap between training and test images is higher. Our normalization technique (see Section 3) is more effective than CLAHE, and of course, the difference is more evident in the Grozi-120 db.

A second experiment has been carried out to understand if and how the color features exploited in the pre-selection phase improve the recognition performance. To this purpose, the recognition has been tested with and without the proposed pre-selection procedure; the AUROC obtained on the Grozi-120 and GP-20 datasets are reported in Table 2 for the BoW- and DNN-based techniques. In all cases, the pre-selection procedure produces positive effects on the recognition accuracy. Furthermore, the number of candidates to be analyzed at the FINE-SELECTION stage is significantly smaller: in Grozi-120 35% of the products can be discarded by pre-selection while in GP-20 the percentage grows to 63%. The values also show that the performance of the two feature representation is very similar on the easier GP-20 dataset, whereas in the Grozi-120 the DNN features offer better performance.

The proposed BoW feature extraction approach has some relevant parameters (i.e. the dictionary size $s$ and the sampling step $p$) and some tests have been conducted to evaluate their influence on the recognition performance. The results obtained are summarized in Table 3 where the AUROC value is reported for different parameter combinations. The results show that the resolution used for the descriptors, i.e. the sampling step, has an important impact on the performance, obviously at the expense of an increment of the computational complexity. On the other hand, the impact of the dictionary cardinality is less evident, and suggests to keep the value of $s$ small thus reducing the overall dimensionality of the feature vector.

As to the sampling step $p$, as expected, the performance of the multi-resolution approach lays in the middle between the two related single-resolution values but in spite of a slight accuracy drop a noticeable efficiency improvement can be reached (the computation time is halved).

In Fig. 11 we show some product examples from the two datasets. For Grozi-120 we show product 1 and product 15 to allow a direct comparison with the results reported in Figs. 7 and 8 of Merler et al. (2007). The AUROC value is reported for each product, together with one training and some testing images. We observe that boxed products, such as product 1, despite of the change of brightness and perspective between test and training images, are characterized by a limited variation in positioning and shape, thus allowing an easier recognition. On the contrary, cans or bottled products (e.g. product 15) which can be placed on the shelf with more degrees of freedom (e.g. rotated) are more difficult to recognize.

These preliminary experiments prove that the proposed features are feasible to the specific problem; as to the BoW features, the multi-resolution approach represents a good trade-off between


**Fig. 15.** Errors during the pre-selection step: (on the left) red box for the ground truth and white dots for points detected by Harris; (center) red dots for the top-left corner of candidates after the pre-selection step and (in the right) the training image.

accuracy and computational complexity and will be used in the subsequent tests.

### 6.2. Evaluation of the detection approach

As suggested in several localization contexts (e.g. Jain & Miller, 2010), the performance of a detection system can be evaluated according to discrete or continuous indicators; in our approach both metrics are therefore considered, and the following indicators are used:

- ***Discrete Metric:*** The performance is measured in terms of Recall and Precision rates. A detected object is considered a True Positive (TP) if its center is inside the ground truth box otherwise it is considered a False Positive (FP). In particular:
  - ***Recall:*** is the percentage of rightly detected objects over the total number of labeled objects (ground truth), namely
  $$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}$$
  - ***Precision:*** is the percentage of rightly detected objects over the total number of detected objects, namely
  $$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$$
  - $FP_{avg}$: is the average number of false positives detected for each processed image.
- ***Continuous Metric:*** The performance is defined in terms of the Overall Recall and Overall Precision rates, as suggested in (Mariano, Min, Park, & Kasturi, 2002). The recall of a ground truth object is the proportion of area covered by the object localized by the method.
  The precision of a detected object is the proportion of its area that covers the ground truth objects. More formally, given a frame $t$:

$$\text{Recall}(t) = \frac{\sum_{G_i}\left|G_i^t \cap \text{U}_{D^t}\right|}{\left|G_i^t\right|} \quad \text{and}$$

$$\text{Precision}(t) = \frac{\sum_{G_i}\left|D_i^t \cap \text{U}_{G^t}\right|}{\left|D_i^t\right|}$$

where $G_i^t$ is the i-th ground truth object in the frame t, $D_i^t$ is the i-th detected object in the frame t, $U_{G^t}$ is the spatial union of the boxes of ground truth objects in t and $U_{D^t}$ the spatial union of the boxes of the detected object in $t$.
Overall Recall and Overall Precision are the weighted average recall and precision over all the frames, respectively:

$$\text{OverallRecall} = \frac{\sum_t \#G^t \cdot \text{Recall}(t)}{\#G^t}$$

and

$$\text{OverallPrecision} = \frac{\sum_t \#D^t \cdot \text{Precision}(t)}{\#D^t}$$

where $\#G^t$is the number of ground truth objects in the frame $t$ and $\#D^t$is the number of detected objects in the frame $t$.

**Table 6**
Product detection: Mean average precision of the proposed approach using the multi-resolution BoW and DNN features on the Grozi-120 and GP-20 datasets.

| *Method* | *Grozi-120* | | *GP-20* | |
|---|---|---|---|---|
| | *Discrete* | *Continuous* | *Discrete* | *Continuous* |
| BoW Multi 12-6 | 0.3431 | 0.2940 | 0.6518 | 0.5848 |
| BoW Multi 6-3 | 0.4435 | 0.3762 | 0.7957 | 0.6821 |
| DNN | 0.4885 | 0.3890 | 0.7574 | 0.6403 |


**Fig. 16.** Errors during the detection steps: (left column) test images and ground truth are shown; (center column) detected objects are red marked; (right column) the training sample is depicted.

Following the same protocol described in (Merler et al., 2007), the detection results obtained on Grozi-120 are shown in order to perform comparisons between our approach and the state of the art. In Table 4 the detection results obtained on the Grozi-120 db are reported both in terms of the continuous and discrete metric. Besides the defined Precision and Recall values, the average number of false positives ($FP_{avg}$) detected in a single image is given; the test considers 10 images containing the target product and 100 images not containing the product, as suggested by the Grozi-120 protocol.

In particular, the performance of the proposed approach (with BoW and DNN features) is compared to the results of the techniques presented in (Merler et al., 2007): CHM (Color Histograms), SIFT (Scale Invariant Feature Transform) and ADA (Adapting Boosting). For the proposed approach, we evaluated the two different features (BoW and DNN) and we compared our post-processing with a typical Non-Maxima Suppression approach. As to the features, in this very complex dataset DNN outperforms BoW reaching higher recall values at a similar level of precision. In both cases, Non-Maxima Suppression produces unsatisfactory results with lower recall values related to the imprecise final shape obtained. An example is given in Fig. 12 where the detection results of the two techniques are compared. Our post-processing allows to better approximate the product shape.

Even if in general the performance obtained on this challenging dataset is not exciting, the results show that the proposed approach largely outperforms the methods reported in (Merler et al., 2007); both the Recall and Precision percentages are about twice those obtained by the other approaches, in both metrics. It is worth noting that the performance over the single products varies significantly, i.e. for some products the detection performance is nearly perfect, while for others no objects are detected. In Section 6.3 we include a more detailed error analysis with the aim to figure out the main causes of failures.

In Fig. 13 we show three examples of object localization performance obtained with our approach (these examples are the same reported in Figs. 10–12 of (Merler et al., 2007)).

Table 5 reports the detection results obtained by the proposed approach on GP-20 db while in Fig. 14 we show three examples of object localization performance. In general the performance on the GP-20 db is significantly better due to the higher quality of the training and testing images. On this dataset BoW features seem to perform slightly better than DNN feature, whose invariance is probably unnecessary for this simpler dataset (see further comments in the following).

For comparison purposes with the state-of-the-art, Table 4 and Table 5 report the results at a fixed threshold value (for the final cluster selection *clThr*, see Section 5); in particular the result with the best trade-off between precision and recall is given. Of course, the relation between the two values varies according to the threshold value; higher thresholds will increase the precision (reducing the number of false positives) at the expenses of a lower recall. To better analyze the overall behavior of the proposed algorithm as a function of the final threshold *clThr* we computed the Mean Average Precision on both datasets, which analyzes the precision as a function of the recall and summarizes the trend reporting the area under the curve; the results are reported in Table 6 where the detection performance of the two multi-resolution approaches (12-6 and 6-3) are compared to DNN features.

The results obtained by the two proposed approaches (BoW and DNN) are very interesting. The DNN features show their superiority on the challenging Grozi-120 db and prove to be more robust to the severe variations characterizing the test images (e.g. blurring or uneven illumination). On the other hand on the high-quality GP-20 db, the BoW approach provides more accurate results and, in particular a lower number of false positives. A visual inspection of the detection results suggests that, in this case, the image partitioning used for BoW feature extraction (see Fig. 7) enforces a strict spatial similarity between training and test images while CNN typical invariance (to translation and occlusions) here leads to a lower accuracy. In fact, most of the false positives found by the DNN-based approach are candidate windows containing portions of two nearby identical products. Such problem is less evident on the Grozi-120 db since many testing images contain a single instance of the target product.

Finally, the efficiency of the proposed approach significantly depends on the setting; considering an average number of 10 training samples for single product (including native and derived poses), processing an image at 3 different scales with the multiresolution 12-6 approach requires about 1 s using non optimized C# code. In our test environment, the complexity of the DNN feature-based algorithm is comparable to that of BoW. We are confident that ad hoc optimizations could significantly reduce the processing time.

### 6.3. Error analysis

Here we discuss the main causes of errors encountered during product detection from shelves, especially on the Grozi-120 db.

- *Pre-selection Error Analysis*: looking at the candidates resulting from the pre-selection procedure we observe that in some cases the right candidates are not detected due to excessive darkness of the product in test images, often blending the object with the background. In Fig. 15 three examples of such errors are shown.
- *Detection Error Analysis*: as to the detection errors, some rather frequent causes of errors can be identified, such as inaccurate ground truth labeling or very large differences between in vitro and in situ images in terms of scale, aspect ratio or color. In Fig. 16 some examples are reported: in the first row we show a false positive corresponding to an unlabeled product which is nearly identical to the labeled one; in the second row a labeling problem (inaccurate box) makes the detected object a false negative (in the discrete metric); in the third row, despite of the different point of view of the object in training and test, the object has been detected in two instances but one is not labeled; from the last row a clear problem of size mismatch between test and training is evident.

## 7. Conclusions

A new approach for detection of products from shelves in unconstrained environments, based on two different feature representations, has been proposed in the present work. The experimental results show the effectiveness of our approach and the improvement given by the color pre-selection step both in term of detection accuracy and efficiency.

Both the feature representations implemented provide quite good accuracy. DNN-based feature are in general more effective in complex scenarios thanks to their high robustness and invariance. BoW-based features with image-partitioning implementation give their best on simpler scenarios where enforcing global structural similarity can be useful.

The post-processing technique adopted also proved its efficacy: a good detection accuracy can be obtained with a very limited number of false positives. The comparison with state-of-the-art techniques on a challenging dataset confirms the validity of our approach.

As a future work, we intend to further reduce the computational complexity of our detection technique. For instance, for the BoW-based approach, one possibility is reducing the number of keypoints considered by retaining only the most salient ones, as recently proposed in Buoncompagni, Maio, Maltoni, and Papi (2015) and Carneiro and Jepson (2009). For DNN-features the complexity can be reduced as well by performing the forward step in all CNN convolutional levels on the whole input image instead of on disjoint candidate sub-windows.

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